SOURCES OF CREDIT RISK

Learning Outcome

- describe credit risk and its components, probability of default and loss given default

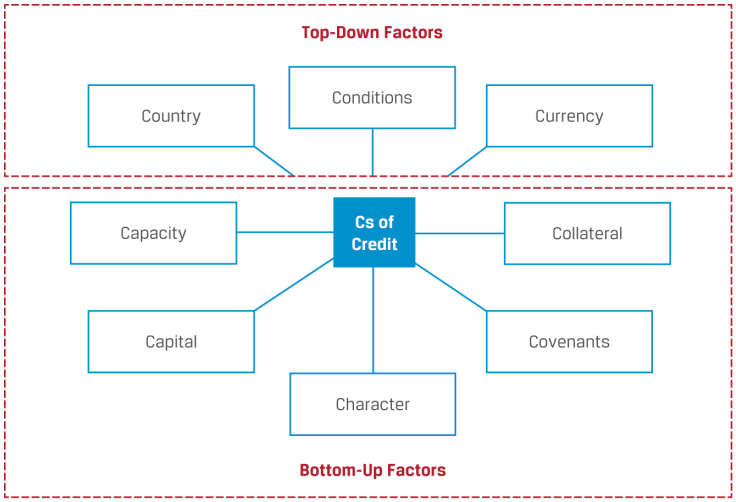
Fixed-income investors face credit risk, a form of performance risk in a contractual relationship. A borrower that fails to meet its promised interest and/or principal payment obligations under a bond or loan contract is said to be in **default**. A fixed-income investor seeks compensation for the expected economic loss under a potential borrower default over the life of the contract known as **credit risk**.

Credit risk depends upon specific factors related to the borrower itself as well as general economic conditions and is subject to change over the life of the contract. Credit risk exposes the lender or investor to potential losses and underperformance.

Traditionally, many analysts evaluated creditworthiness based on what are often called the "Cs of credit analysis," as shown in Exhibit 1:

Exhibit 1:

The Cs of Credit Analysis



▶ Expand / collapse Extended Description

Five of these criteria—capacity, capital, collateral, covenants, and character—are related to the specific bottom-up factors applicable to an individual borrower. **Capacity** refers to the ability of the borrower to make its debt payments on time. **Capital** addresses other company resources available that reduce reliance on debt. **Collateral** refers to the quality and value of the assets supporting the issuer's indebtedness, while **covenants** are the legal terms of debt agreements that an issuer must comply with. **Character** refers to the quality of management and the willingness of repay indebtedness.

While capacity and capital are generally quantitative metrics based on financial statements, collateral, covenants, and character are largely qualitative measures that analysts assess based on historical company performance, credit relationships and the reputation of current management.

general economic, competitive, and business environment faced by all borrowers that may affect their ability to service or refinance debt. **Country** involves the geopolitical environment as well as the legal and political system faced by all issuers in a jurisdiction that may affect debt payment. **Currency** affects issuers whose cash flows are affected by exchange rate changes or who borrow in a currency outside of their jurisdiction, such as sovereign issuers with foreign currency debt.

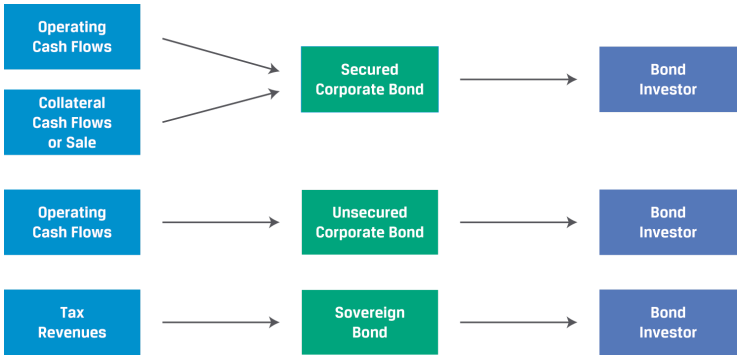
A borrower's inability to make timely payments may be due to several underlying and contributing factors, but it ultimately results from a lack of sufficient cash available to make a current debt payment. Lenders, as fixed-income investors, are impacted by credit risk. They may lose some or all of the principal and interest due, suffering a capital loss or income shortfall. This disruption to their cash flows may impact their own ability to repay debt, and payment delays as well as increased legal and collection costs may further reduce their income and margins.

Sources of Credit Risk

Changes in credit risk due to issuer-based or overall economic conditions are affected by the source of repayment, which differs among issuer types. Exhibit 2 shows the main sources of repayment for corporate and sovereign bonds.

Exhibit 2:

Sovereign and Corporate Bond Sources of Repayment



▶ Expand / collapse Extended Description

A corporation may not generate sufficient cash flow from its operations, financing activities, and investments to make timely payments on its debt, increasing the risk of default. Similarly, a sovereign borrower may not generate sufficient revenue through taxes, incomes, and fees to make timely payments.

Borrowers that lack the resources to meet debt obligations as they come due are often described as illiquid. An illiquid borrower is unable to raise the necessary funds to fulfill a debt obligation. They may not be able to tap credit lines, sell assets, or otherwise raise funding to make a timely debt payment. That is different from an *insolvent* borrower whose assets are worth less than its liabilities. An illiquid borrower just does not have, or is unable to secure, the liquidity needed to make timely payment.

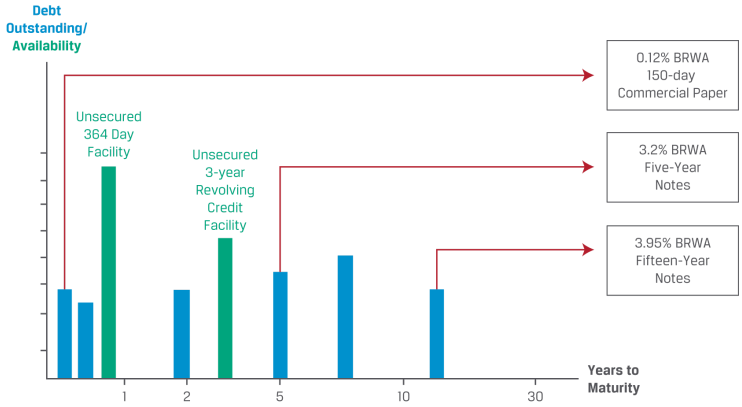
Credit risk analysis focuses on understanding how a borrower generates cash, how and when cash is used, and what events and risks may impede the borrower's ability to repay the debt. In earlier lessons, we encountered *unsecured* debt, where the primary source of repayment is the cash flow generated by the business. In the case of *secured* debt, the primary source of repayment remains the firm's cash flows, but it is also supported by collateral pledged by the company. The following example involves an investor considering an unsecured bond versus a secured bond.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

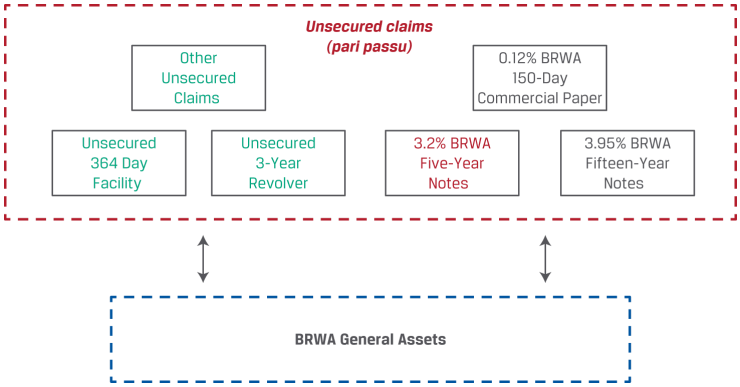
Risk Comparison of BRWA and VIVU Notes

The hypothetical car manufacturer Bright Wheel Automotive (BRWA) is an investment-grade (IG) issuer with ready access to unsecured debt, while Vivivyu Inc. (VIVU) is a hypothetical high-yield (HY) issuer that borrows on a secured basis. Let us compare 3.2% senior unsecured and unsubordinated BRWA notes to 6.5% secured unsubordinated VIVU notes. Consider BRWA's debt profile:

Figure 1: BRWA Corporation Debt Profile

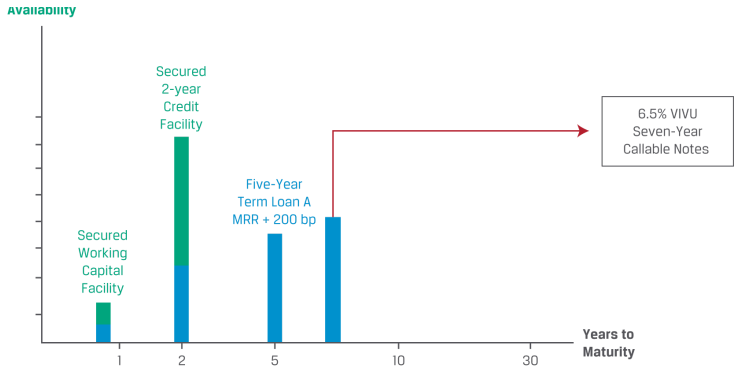


If BRWA fails to make an interest or principal payment on the note, BRWA is in default. Since its bonds are unsecured and unsubordinated obligations with pari passu and cross-default language in its bond indenture, failure to repay unsecured indebtedness means that all its debt obligations are in default and investors must rely on BRWA's general asset pool to satisfy their obligations, as shown below.



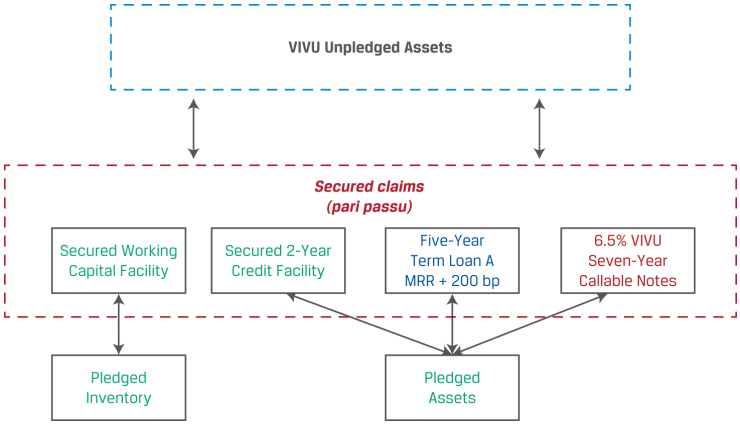
As an HY issuer, on the other hand, VIVU is limited to secured debt market access, as shown in its debt profile.

Figure 2: VIVU Debt Profile



Note: MRR stands for market reference rate.

VIVU's failure to make timely payments due on the note will also result in default. The pari passu clause and cross-default language in its bond indenture with other unsecured indebtedness also sends all VIVU debt obligations into default. While *all* of VIVU's debt investors have access to any unpledged assets, specific pledged assets serve as a secondary source of repayment for specific debt, as shown below.



In this example, VIVU's secured claims include its working capital facility secured by short-term assets (in this case inventory), while its long-term secured debt including the callable notes are backed by a claim to specific long-term assets pledged. As a secondary source of repayment, only when the value of the pledged assets falls below the total amount of pari passu secured debt will VIVU noteholders begin to suffer a credit loss.

While the *likelihood of default* for investment-grade borrowers, such as BRWA, is typically well below that of a high-yield issuer such as VIVU, the investor's *loss in the event of a default* for secured debt is usually lower than unsecured debt due to the secondary source of repayment for secured bonds.

We encountered the Government of Romania's EUR1.95 billion 30-year debt issue in an earlier lesson. For a sovereign government, the right to tax economic activity within its borders means that the sources of repayment are the personal and corporate income taxes, sales taxes, and other revenue generated by the government. In the specific case of Romania's EUR-denominated debt, the government's ability to generate reserve currency cash flows based upon its institutional and economic profile, use of monetary and fiscal policy, and exchange rate regime are additional considerations.

The main borrower types and their primary sources of repayment and credit risk are listed in Exhibit 3.

Borrower Types, Sources of Repayment, and Sources of Credit Risk

Borrower Type	Primary Source of Cash Flow Generation	Secondary Source of Cash Flow Generation	Sources of Credit Risk
Corporate	▪ Business operations ▪ Investment and financing activities	▪ Asset sales ▪ Divestitures ▪ Additional debt issuance ▪ Equity issuance	▪ Economic contraction ▪ Strategic shifts in the business and market environment ▪ Increased competition ▪ Reduced pricing power ▪ Shrinking operating margins, increased losses ▪ Excessive debt service needs
Sovereign or public entity	▪ Corporate and personal income taxes, sales tax and VAT revenue, capital gains and wealth-based taxes ▪ Tariffs, fees, and other government revenue	▪ Newly issued debt ▪ Sale of public assets, privatization	▪ Economic contraction ▪ Political uncertainty ▪ Excessive debt service needs ▪ Expansionary economic policies ▪ Budget deficits ▪ Tax cuts ▪ Limited ability to collect taxes

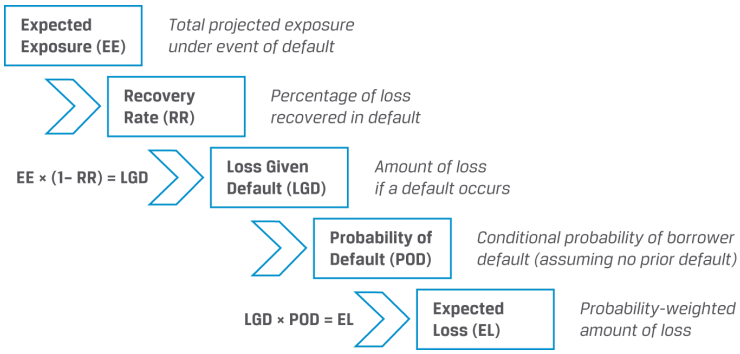
While many factors affect the primary and secondary sources of repayment for different fixed-income issuers, investors seek to measure and compare credit risk among different borrowers for similar maturities as well as across maturities for the same borrower.

For example, consider the comparison of BRWA and VIVU notes in Example 1. If we set aside both the VIVU maturity difference and call feature (which the issuer may exercise any time from year three through maturity in year seven) by assuming a similar expected maturity to the BRWA bond for the time being, compare the 6.5% VIVU debt coupon to the 3.2% fixed rate of the BRWA bond. How can we assess whether the additional 3.3% (= 6.5% − 3.2%) in interest adequately compensates an investor for the additional risk? We turn to measures of credit risk, which help answer this question in the next section.

Measuring Credit Risk

Fixed-income investors face credit risk, or the risk that an issuer does not make promised interest and principal payments. This probability-weighted shortfall known as the **expected loss (EL)** for a given period has two components, as shown in Exhibit 4.

Components of Credit Risk



▶ Expand / collapse Extended Description

The first is known as the **probability of default (POD)**, which is the likelihood that an issuer fails to make full and timely payments of principal and interest. The probability of default is typically an annualized measure.

The second component is the **loss given default (LGD)**, or the investor's loss conditional on an issuer event of default. The loss given default combines the severity of loss under a default scenario with the amount of the investor's claim at the time of default. The **recovery rate (RR)** represents the percentage of an outstanding debt claim recovered when an issuer defaults, while the **loss severity** of (1 − RR) represents the unrecovered portion of the claim. The size of the investor's claim at the time of default is known as the **expected exposure (EE)** or **exposure at default (EAD)**. This is the amount an investor may expect to lose in the case of default, which is usually equal to the loan or bond face value plus accrued interest less the current market value of available collateral. In most instances, in the event of default, bondholders will usually recover some proportion of their investment.

The expected loss is shown in Equation 1:

$$EL = POD \times LGD, \qquad (1)$$

where LGD = EE × (1 – RR).

The LGD can be expressed either in currency terms (e.g., EUR450,000) or as a percentage of principal (e.g., 45%). The latter form of expression is more useful for analysis since it allows comparison across borrowers and investments of different sizes.

One way to interpret the expected loss on a fixed-income security for a given period is to compare it to the compensation an investor expects for taking on the credit risk of a borrower over that period, which is the credit spread. Recall from an earlier lesson that the G-spread is equal to yield spread in basis points over an actual or interpolated government bond. This approximation may be expressed as follows:

$$\text{Credit Spread} \approx POD \times LGD. \qquad (2)$$

Therefore, we can say that the investor is fairly compensated if the expected loss is equal to the credit spread for a given period. The following example uses this information to assess the expected loss versus compensation for the BRWA and VIVU bonds from Example 1.

EXAMPLE 2

Comparing Expected Loss and Credit Spreads

Recall from Example 1 that Bright Wheel Automotive (BRWA) issued an unsecured five-year bond with a 3.2% fixed coupon. The yield-to-maturity difference between BRWA's five-year bond and the comparable US Treasury bond is 90 bps (= 3.2% –

call at a fixed price any time from three years from the issuance date until final maturity in seven years.

1. Assuming the BRWA bond has a probability of default of 1% and a loss given default of 80%, estimate whether BRWA bond investors are adequately compensated for assuming BRWA credit risk.

   Hide Solution

   Solution:

   Using Equation 1, we may solve for BRWA's expected loss in percentage terms as follows:

   EL = POD × LGD, or 0.8% = [0.01 × 80%].

   Given BRWA's credit spread of 0.9%, we may conclude based upon the approximation in Equation 2 that BRWA investors are fairly compensated for assuming the credit risk since:

   Credit Spread > POD × LGD.

   Investors earn a 90 bps spread per year and have an expected loss of 80 bps per year.

2. If we assume that VIVU has issued a five-year non-callable bond, as in the case of BRWA, with a probability of default of 6% and a loss given default of 50%, compare investor compensation for assuming the credit risk of the 6.5% VIVU debt coupon to that of the 3.2% fixed rate of the BRWA bond.

   Hide Solution

   Solution:

   Using Equation 1, we may solve for VIVU's expected loss in percentage terms as follows:

   EL = POD × LGD, or 3.0% = [0.06 × 50%].

   We may calculate VIVU's five-year credit spread by subtracting its 6.5% coupon from the 2.3% five-year US Treasury to solve for a G-spread of 4.2%. Based upon the approximation in Equation 2, we may conclude that VIVU investors are fairly compensated for assuming the credit risk since:

   Credit Spread > POD × LGD.

   Investors earn a 420 bps per year spread and have an expected loss of 300 bps per year.

3. How would the determination of the spread as well as investor's view of the risk versus compensation for the VIVU bond change from Question 2 if we consider the VIVU bond's *actual* maturity and call feature with the original 6.5% coupon?

   Hide Solution

   Solution:

   VIVU investors face both call risk and greater maturity risk on the original bond terms versus a five-year, non-callable bond. As described in earlier lessons, bonds callable at a fixed price prior to maturity are advantageous to issuers and disadvantageous to investors, so investors expect to be compensated via a higher yield versus a similar non-callable bond given the uncertain maturity and limited price appreciation known as call risk. This option-adjusted yield is the required market discount rate for which the bond's price is adjusted for the value of the embedded option. Investors would also

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

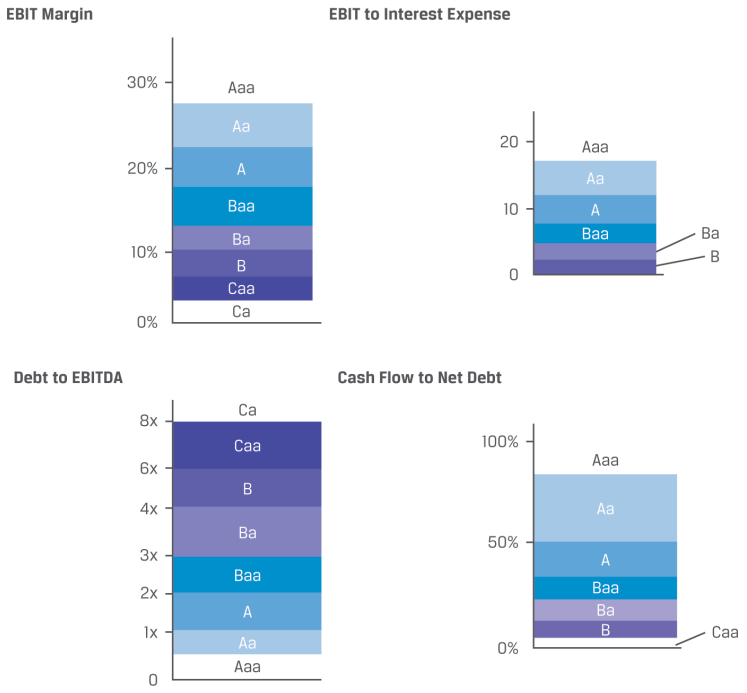
We may further consider the drivers of expected loss for BRWA's unsecured investment-grade debt as compared to VIVU's secured high-yield debt in Example 2. The POD is driven by an issuer's ability to service debt based on both qualitative and quantitative factors, including:

- Profitability: Stable, predictable cash flows and profits;
- Coverage: Sufficient cash flows/profits to make debt payments; and
- Leverage: Relative reliance on debt financing.

Higher profitability (as measured, for example, by EBIT margin) and higher coverage (EBIT to interest expense) coupled with lower leverage (debt to earnings or cash flow ratios, such as debt/EBITDA or cash flow to net debt) are associated with lower POD and higher credit quality. While financial ratios are one of several factors used to assess credit risk, and their relative level by rating category varies widely across industries and over time, Exhibit 5 provides a generic example of these relationships.

Exhibit 5:

Key Financial Ratios for Corporate Credit Risk and Ratings

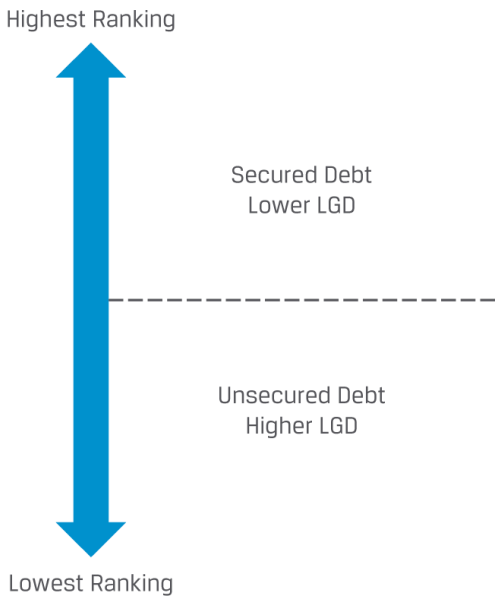


▶ Expand / collapse Extended Description

As Exhibit 5 shows, changes in profitability (EBIT margin), coverage (EBIT to interest expense), and leverage (debt/EBITDA and cash flow to net debt) ratios over time may result in ratings upgrades or downgrades as well as changing credit spreads. The LGD for a given exposure, on the other hand, is largely a function of the nature and seniority of a creditor's claim in a default scenario, as shown in Exhibit 6:

Exhibit 6:

Seniority and LGD



▶ Expand / collapse Extended Description

For investors in unsecured investment-grade bonds or loans with a high LGD, the greatest risk of expected loss arises due to a rise in POD. On the other hand, high-yield investors expecting a greater likelihood of default seek to minimize EL by seeking covenant restrictions and/or security to lower LGD.

Question Set

1. *Determine the correct answers to fill in the blanks:* The two components of credit risk are the _____, typically measured over a 12-month horizon, and the _____, which together with the recovery rate determine the _____.

Hide Solution

Solution:

The two components of credit risk are the **probability of default**, typically measured over a 12-month horizon, and the **exposure at default (or expected exposure)**, which together with the recovery rate determine the **loss given default**.

2. Which of the following statements best describes the difference between unsecured and secured debt obligations in the event of issuer default?

A. Unsecured debtholders only have access to issuer cash flows, while secured debtholders only have access to specific pledged assets.
B. Both unsecured and secured debtholders have access to unpledged assets, while only secured debtholders have access to specific pledged assets as a secondary repayment source.
C. Only unsecured debtholders have access to pledged assets, while only secured debtholders have access to specific pledged assets.

Hide Solution

Solution:

B is correct. While all debtholders receive interest and principal payments from issuer cash flows, in an event of default both unsecured and secured debtholders have access to unpledged assets, while only secured debtholders have access to specific pledged assets as an additional repayment source.

A. Higher profitability, higher coverage, and higher leverage
B. Higher profitability, lower coverage, and lower leverage
C. Higher profitability, higher coverage, and lower leverage

Hide Solution

Solution:

C is correct. Higher profitability, higher coverage, and lower leverage are associated with a lower probability of default and higher credit quality for a corporate issuer.

4. A bond investor analyzing Broadvue Corporation's unsecured debt estimates a POD of 2% and an LGD of 80%. Using this information to approximate the annual credit spread and observing an actual credit spread of 200 bps per year, which of the following statements is correct?

A. An investor would be fairly compensated for assuming Broadvue's credit risk.
B. An investor would be less than fairly compensated for assuming Broadvue's credit risk.
C. An investor would be more than fairly compensated for assuming Broadvue's credit risk.

Hide Solution

Solution:

C is correct. Since POD × LGD = 1.60% and the credit spread is 2.00%, Credit Spread > POD × LGD and the investor would expect to be more than fairly compensated for assuming Broadvue's credit risk.

Discuss Filter

Discussion

What do you want to discuss... (required)

JG

In example 2, question 2: "Investors earn a 420 bps per year spread and have an expected loss of 300 bps per year." and solution says they are "fairly compensated". Thats 40% spread more than credit risk (or 120bps). On the other hand in the question set Q4 for 200 spread vs 160 risk (25% and 40 bps) solution Fairly is stated as incorrect because thats apparently more than fairly compensated. Math isnt mathing here CFA...

Created 3 months ago by Jakov Gvozden

0 replies | Last Activity: 3 months ago

Reply to this Comment

© 2025 CFA Institute. All Rights Reserved.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search